Tertiary extension and tilting in the Queen Charlotte Islands, evidence from dyke swarms and their paleomagnetism[1]

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The Queen Charlotte Islands form the western margin of the Tertiary Queen Charlotte Basin, which is situated on the western margin of the North American Plate. They contain seven major dyke swarms of Late Eocene to Miocene age, a period when the relative motions of the Pacific and the North American plates in this region were dominantly dextral strike slip (transform margin), with intervals of highly oblique divergence and convergence. Within each swarm, dykes have a systematic trend. However, trends vary from swarm to swarm, indicating that the stress field varied. A total of 678 cores (1352 specimens) were collected from 129 dykes in six swarms over a distance of about 200 km. Magnetic stability is variable. One hundred and one dykes yielded records of the paleofield. Data are also reported from an Oligocene pluton (5 sites, 27 cores, 52 specimens) and Miocene lavas (8 sites, 52 cores, 101 specimens). Both normal and reversed magnetizations occur, but irrespective of sign, the mean directions of remanent magnetization of each swarm and of the pluton and the lavas have systematically steeper inclinations than the value calculated from coeval rocks in North America. To explain this it is proposed that, after dyke emplacement, the sampling areas were tilted to the north or northwest by amounts that vary between 9 and 16°. Apparently, crustal tilting, similar in magnitude and direction, extended over distances of approximately 200 km. This cannot reflect tilting of a single block. Instead, it is argued that at least the southern Queen Charlotte Islands underwent considerable northerly or north-northwesterly directed extension and normal block faulting, which followed and in part was concurrent with the formation of widespread mid-Tertiary dyke swarms, plutons and lava flows. Making use of the fact that dykes propagate perpendicular to the direction of extension, and combining previously measured dyke orientations with paleomagnetic data, three stages of extension are proposed: east–west extension sometime during the Late Eocene to Early Oligocene; north–south extension sometime in the interval Late Oligocene to Early Miocene; and northwest–southeast extension sometime during Late Miocene or later time.

Les îles de la Reine-Charlotte forment la marge occidentale du bassin de la Reine-Charlotte, d'âge tertiaire, lequel est localisé sur la marge occidentale de la plaque de l'Amérique du Nord. Elles comportent sept essaims de dykes majeurs, âgés de l'Éocène tardif à Miocène, période durant laquelle les déplacements relatifs des plaques du Pacifique et de l'Amérique du Nord dans cette région s'effectuaient par des décrochements dextres (marge transformante), incluant des intervalles de divergence et de convergence largement obliques. Chaque essaim est caractérisé par des dykes avec une orientation systématique. Cependant, les orientations varient d'un essaim à un autre, ce qui indique des changements dans le champ des contraintes. Au total, 678 carottes (1352 spécimens) ont été échantillonnées sur 129 dykes dans six essaims et sur une distance d'environ 200 km. La stabilité magnétique est variable. Les données du paléochamp ont été obtenues pour 101 dykes. En outre, des données sont présentées pour un pluton oligocène (5 sites, 27 carottes, 52 spécimens) et des laves miocènes (8 sites, 52 carottes, 101 spécimens). Des aimantations normale et inverse ont été fournies, mais indépendamment de leur signe, les directions moyennes de l'aimantation rémanente pour chaque essaim, et pour le pluton et les laves, ont des inclinaisons systématiquement plus abruptes que la valeur calculée pour les autres roches de même âge en Amérique du Nord. Afin d'expliquer cette différence, il est proposé que dans les régions échantillonnées, ultérieurement à la mise en place, les formations géologiques furent basculées vers le nord ou le nord-ouest par des angles compris entre 9 et 16°. Apparemment, ce basculement crustal, avec la même rotation et direction, s'est prolongé sur des distances d'environ 200 km. Donc, il ne s'agit pas d'un seul bloc. Il semble plutôt que les îles de la Reine-Charlotte, au moins dans leur partie sud, furent considérablement affectées par des évènements d'extension et de morcellement par des failles normales de direction nord ou nord-nord-ouest, qui se sont produits après, et en partie concurremment avec, la formation des essaims de dykes régionaux, des plutons et des effusions de laves, datant du Tertiaire moyen. Considérant le fait que les dykes se propagent perpendiculairement à la direction d'extension, et en combinant les mesures déjà connues des orientations des dykes avec les données paléomagnétiques, trois stades d'extension sont proposés : une extension est–ouest pendant l'intervalle Éocène tardif à Oligocène précoce; une extension nord–sud pendant l'intervalle Oligocène tardif à Éocène précoce; et une extension nord-ouest–sud-est pendant le Miocène tardif ou ultérieurement.

[Traduit par la rédaction]

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Introduction

The Queen Charlotte Islands form the western subaerial margin of the Queen Charlotte Basin. The latter is considered to be a Tertiary extensional basin, now mainly below sea level, situated on the northwest margin of the North American Plate (Fig. 1). The islands were a major centre of Tertiary igneous activity that produced thick lava sequences, numerous plutons, and dyke swarms over an interval of about 35 Ma, extending from Late Eocene through Miocene time. It is with the latter, and the record of crustal extension that they provide,

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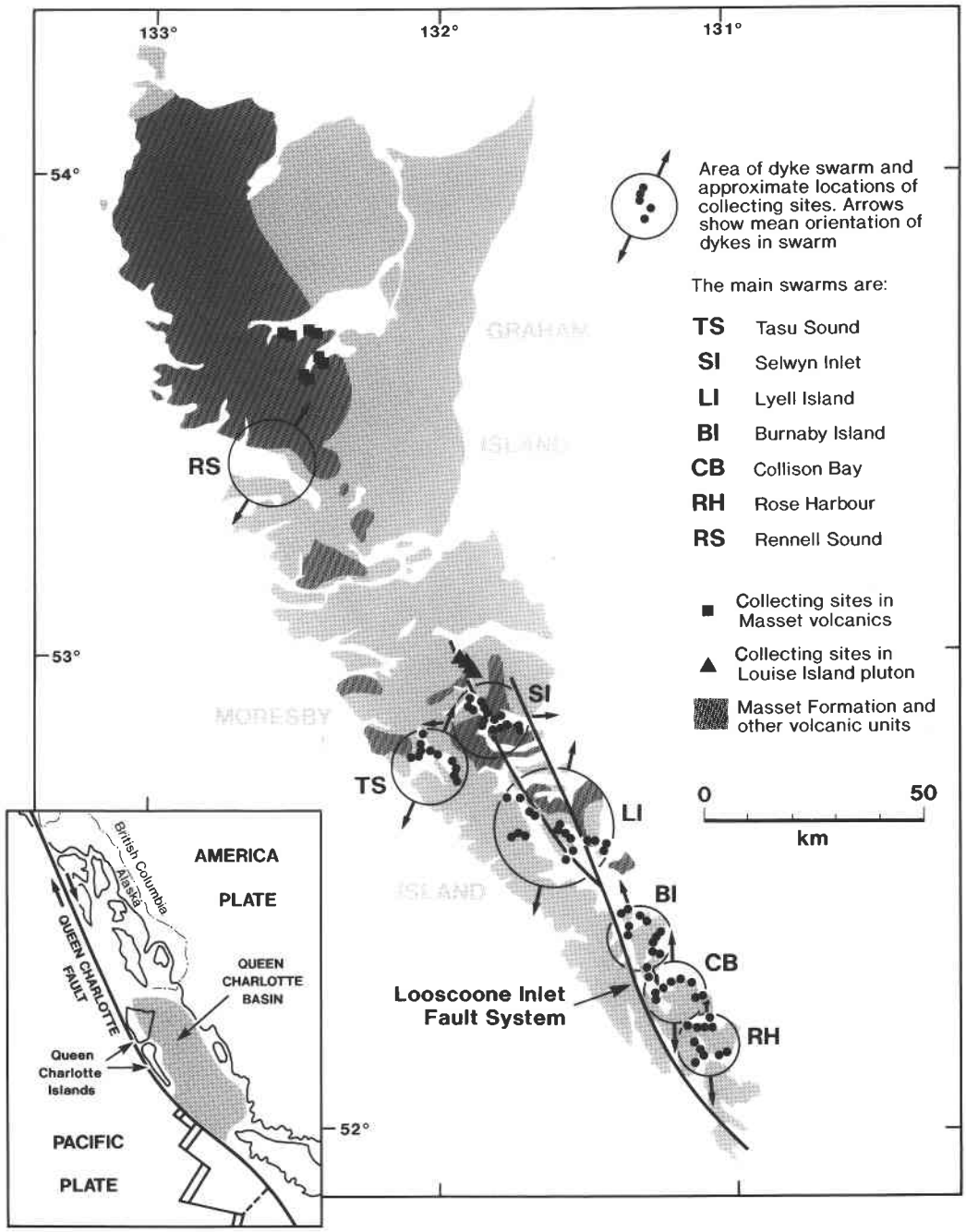


FIG. 1. General distribution of Tertiary igneous rocks of the Queen Charlotte Islands and paleomagnetic sampling sites (precise locations given in Table 3). Present tectonic setting shown in the inset.

that we are principally concerned in this paper.

The hypothesis that the Queen Charlotte Basin is an extensional basin, originally proposed by Yorath and Chase (1981), has received strong support from offshore seismic refraction studies (Rohr and Dietrich 1991). That the islands themselves also have been strongly affected by extensional tectonics is indicated by several lines of evidence, notably the mapping of dyke swarms (Souther and Jessop 1991) and of northerly tilted fault blocks in eastern Moresby Island (Lewis 1991), and the studies of Hyndman and Hamilton (1991), which indicate that the igneous rocks are of a type characteristic of rifted continental margins. The latter authors suggest that extension was between 10 and 40%, and they relate the onset of extension and major volcanism and intrusion to major plate reorganization during the Late Eocene (43 Ma) when convergence of the Pacific and North American plates, which had operated prior to that time, changed to the present dominantly dextral strike-slip motion (Fig. 1). Since 43 Ma, however, the reconstructions of Stock and Molnar (1988) indicate that strike-slip motion was interrupted by intervals of highly oblique convergence and divergence.

In 1987 a systematic assessment of the hydrocarbon potential of Queen Charlotte Basin was begun by the Geological Survey of Canada. Dyke studies were undertaken in 1987 and 1988 to estimate the thermal impact of dyke emplacement on hydrocarbon maturation (Souther and Jessop 1991). Seven swarms of subvertical dykes were identified. The orientation of dykes within each swarm fall within a narrow range, but the

TABLE 1. K–Ar dates from hornblende in Queen Charlotte Islands dykes

Sample	Swarm	K (wt.%)	Radiogenic ^{40}Ar (10^{-7} cm^3/g)	% atmospheric ^{40}Ar	Age (Ma) ($\pm 2\sigma$)
AT-87-7-3[a]	RH	0.511	8.783	34.0	43.7±1.1
SE042387[b]	CB	0.241	2.894	71.0	41.4±4.6
SE041387[b]	CB	0.217	4.669	41.0	54.5±3.2
SE012688	SI	0.169	0.8571	88.0	13±5.7

NOTES: All determinations by J. C. Roddick, Geological Survey of Canada, Ottawa. See Table 4 for swarm abbreviations.
[a]Anderson and Reichenbach (1991).
[b]Souther and Jessop (1991).



FIG. 2. North-trending basalt dyke of the Burnaby Island dyke swarm cutting Triassic Kunga limestone in Skincuttle Inlet.

mean direction varies from swarm to swarm (Fig. 1).

Dykes record the direction of extension at the time of emplacement, and the differences in the trends of swarms may be explained in several ways. Souther and Jessop (1991) proposed that the islands comprise a complex of small tectonic domains separated by major faults, each domain having a different stress field. In contrast, Higgs (1990, 1991), to explain widely varying paleocurrent measurements in sandstones of the Cretaceous Honna Formation, speculated that small blocks with dimensions of the order of 10 km have undergone relative rotation about vertical axes. Hicken and Irving (1977), on the basis of paleomagnetic data, also invoked rotation of central Graham Island. If such rotations are characteristic of the Queen Charlotte Islands as a whole, then the differences in swarm orientation observed by Souther and Jessop could have been the result of postemplacement rotation around vertical axes. However, the fact that the Tertiary igneous episode extended over a period when plate motions in this region were changing (Engebretson *et al.* 1985; Stock and Molnar 1988), suggests that differing dyke orientations may reflect changes in the stress field with time rather than tectonic rotations.

The present study was designed to test these ideas paleomagnetically. Earlier paleomagnetic studies have been reported from lavas around Masset Inlet and a small pluton on Louise Island (Hicken and Irving 1977) and from bore core in Tertiary lavas on Graham Island (Wynne and Hamilton 1989) to the north of our sampling area.

General geology

Regional geology

The Queen Charlotte Islands, with a combined area of 10 000 km^2, are bounded on the west by the Queen Charlotte dextral transform fault, beyond which is the oceanic crust of the Pacific Plate (Fig. 1). The oldest rocks known are Permian sediments (Hesthammer *et al.* 1991), but the Upper Triassic Karmutsen Formation is the oldest unit exposed on most islands. The latter forms a massive platform, about 4000 m thick, of tholeiitic basalt and marine sediments overlain by thinner sedimentary and volcanic successions of Late Triassic to Late Cretaceous age (Sutherland Brown 1968). The degree and style of deformation varies from place to place, suggesting that the islands comprise a mosaic of discrete structural domains separated by major, long-lived faults (Thompson *et al.* 1991). All older rocks are cut by Tertiary plutons and dyke swarms and overlain unconformably by Tertiary volcanics. In the northeast, older rocks are largely hidden beneath an overlap assemblage of Tertiary and Quaternary marine sediments and minor volcanic rocks, which occupy the centre of the Queen Charlotte

TABLE 2. Distribution of types of magnetizations in dyke swarms

Swarm	*B*	*N*	*R*	*N* + *R*	BO	*L*	HD
Tasu	20	4	7	3	1	0	5
Selwyn	24	7	6	6	3	0	2
Lyell Island (including Bigsby)	25	8	10	0	5	2	0
Burnaby Island	20	1	14	2	2	0	1
Collison Bay	20	8	10	0	2	0	0
Rose Harbour	20	4	10	1	1	0	4
Totals	129	32	57	12	14	2	12

NOTES: *B*, number of dykes sampled; *N*, number of dykes with normal polarity; *R*, number of dykes with reversed polarity; BO, number of dykes with Brunhes overprint only; L, number of dykes affected by lightning; HD, number of dykes without coherent magnetizations.

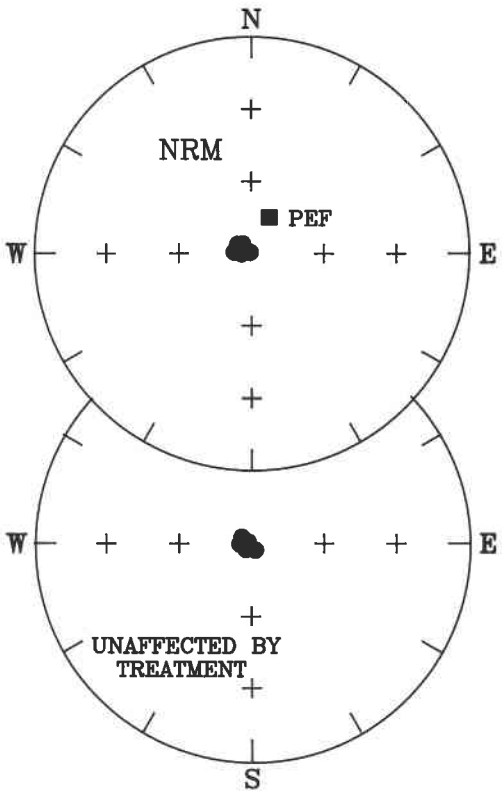


FIG. 3. Directions of magnetization in 10 specimens cut from five oriented cores taken from dyke 17 of the Collison Bay swarm, an example of single-component normal magnetization. Above, the initial (NRM) directions. Below, directions after demagnetization at 500°C and 10 mT (Table 3). PEF is the direction of the present Earth's field. The perimeter is the present horizontal plane. The symbols are all solid, indicating directions plotted on the lower hemisphere (normal polarity).

Basin and which are largely concealed beneath the waters of Hecate Strait.

Tertiary geology

Tertiary magmatic rocks include the Late Eocene to Oligocene Kano Plutonic Suite, an unnamed group of Paleocene to Lower Miocene volcanic rocks, and the Late Oligocene to Early Pliocene Masset Formation (Fig. 1). Dykes, believed to be comagmatic with the plutons and volcanic rocks, are widely distributed throughout the islands, especially in the south.

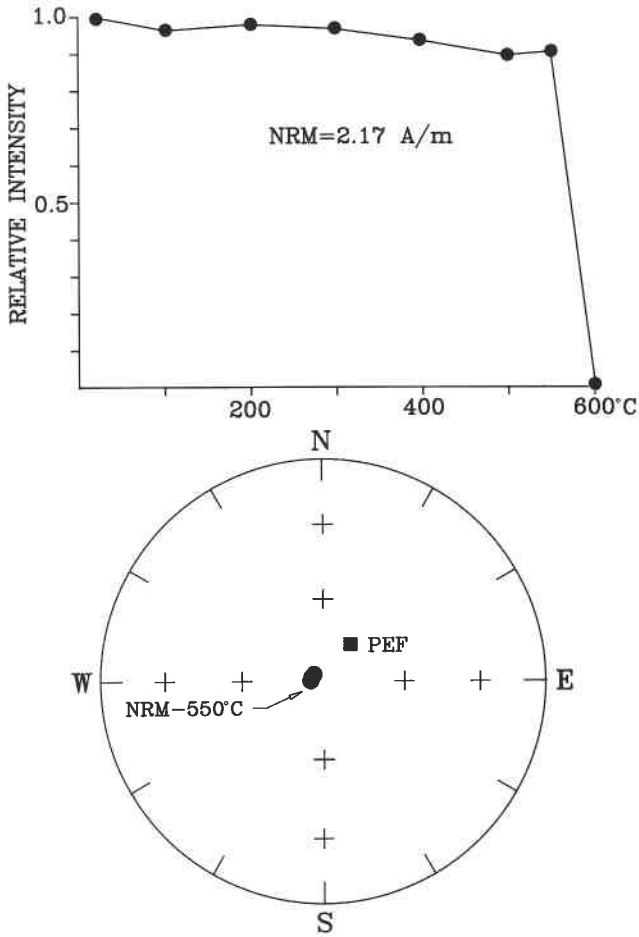


FIG. 4. Thermal demagnetization of specimen from dyke 17 of the Collison Bay swarm. Above, the decay of intensity with temperature; below, the corresponding direction changes. See Fig. 3 for all directions observed from this dyke.

Tertiary volcanic rocks cover much of Graham Island, and remnants are exposed on northern Moresby and adjacent islands. They range in composition from basalt through andesite and dacite to rhyolite. All were formerly included in the Masset Formation (Sutherland Brown 1968), but Hickson (1988) now has suggested that the name Masset Formation be restricted to volcanic rocks of Late Oligocene to Late Miocene age (ca. 30 to 15 Ma). K–Ar dates on volcanic rocks range from less than 5 to 45.7 Ma (summarized in Hickson 1991).

The Kano Plutonic Suite includes at least 11 plutons (Anderson and Reichenbach 1991). On the basis of U–Pb chronometry, three age group have been recognized: a southern Eocene group on southern Moresby Island (46–39 Ma); a medial Early Oligocene group (36–32 Ma); and a northern group of Late Oligocene age from Graham Island (28–27 Ma).

The style and intensity of Tertiary deformation varies from place to place. The Tertiary volcanic rocks on Graham Island show slight tilting and minor normal faulting (Hickson 1991). In contrast, Thompson and Thorkelson (1989) report intense post-Cretaceous folding and faulting along the Rennell Sound fold belt, which extends northwesterly across northern Moresby and southern Graham islands. South of the Rennell Sound fold belt and east of the Louscoone Inlet Fault, Lewis (1991) has mapped northerly tilted fault blocks that reflect extension related to post-Oligocene dextral strike slip.

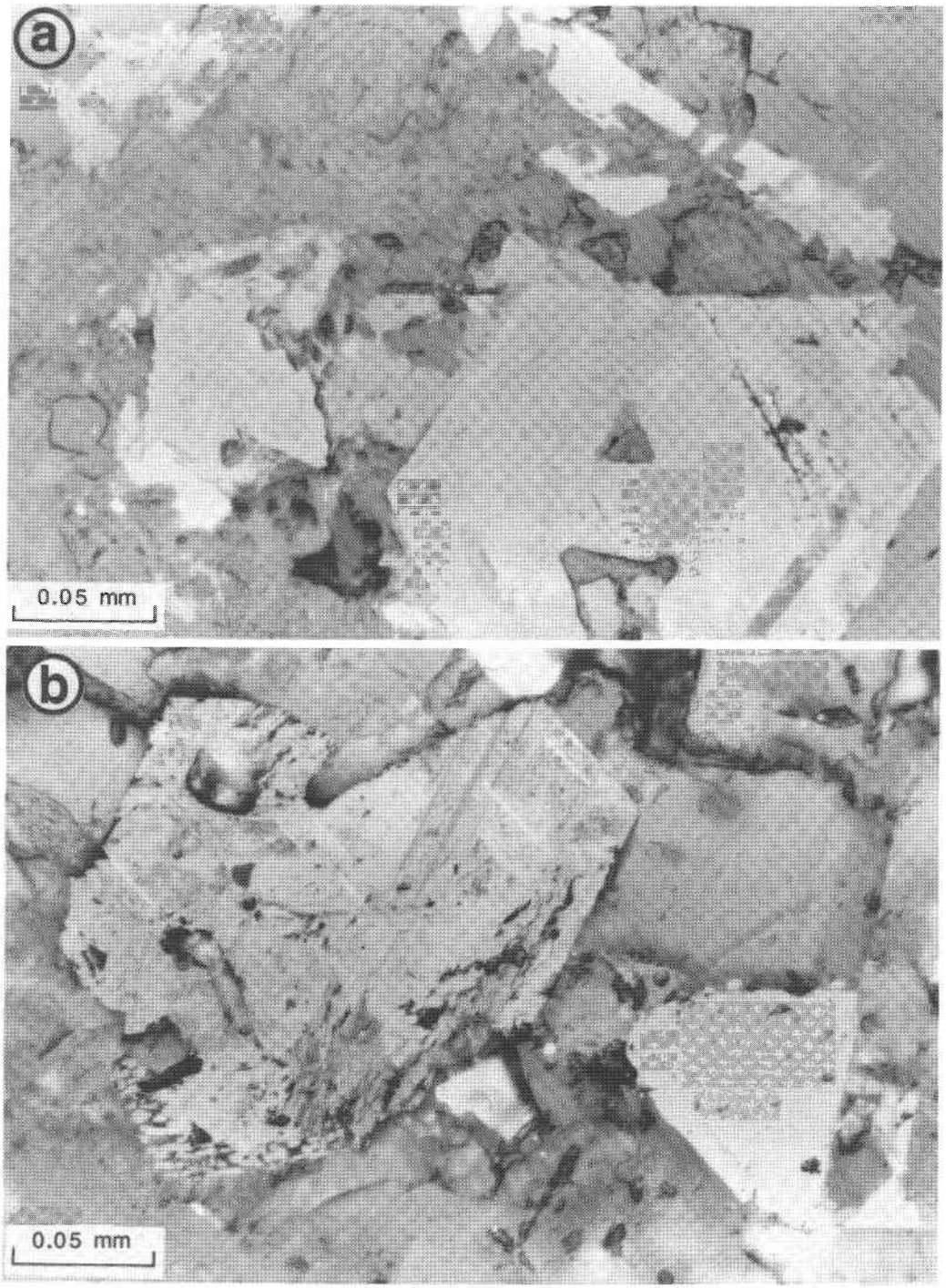


FIG. 5. Reflected-light photomicrographs taken in polarized light of polished sections showing the minerals and textures associated with different magnetization types. (*a*) Collison Bay swarm, dyke 17, basalt with single normal vertical magnetization (Fig. 4); large euhedral grain of titaniferous magnetite with exsolution structure; elongated grains of associated ilmenite (high reflectance) have no visible internal alteration or exsolution textures. (*b*) Tasu Sound swarm, dyke 19, andesite with stable reversed magnetization and Brunhes overprint; euhedral grain of titaniferous magnetite with patches of moss-like alteration and complex exsolution lamellae of ilmenite, reflective grain in lower right is uniform unaltered ilmenite (Figs. 6, 7).

Tertiary dyke swarms and their sampling

General

There are six dyke swarms on southern Moresby and adjacent islands and a major swarm in the Rennell Sound area of southeastern Graham Island (Fig. 1; Souther and Jessop 1991). Minor changes have been made to swarm boundaries as originally mapped. Dykes in Bigsby Inlet have the same magnetic directions as those on adjacent Lyell Island. Thus, the name Bigsby Inlet swarm is abandoned, and data from it included in the Lyell Island swarm. Because of the large number of dykes in the original Carpenter Bay swarm, it is subdivided into the Collison Bay and Rose Harbour swarms. The Rennell Sound

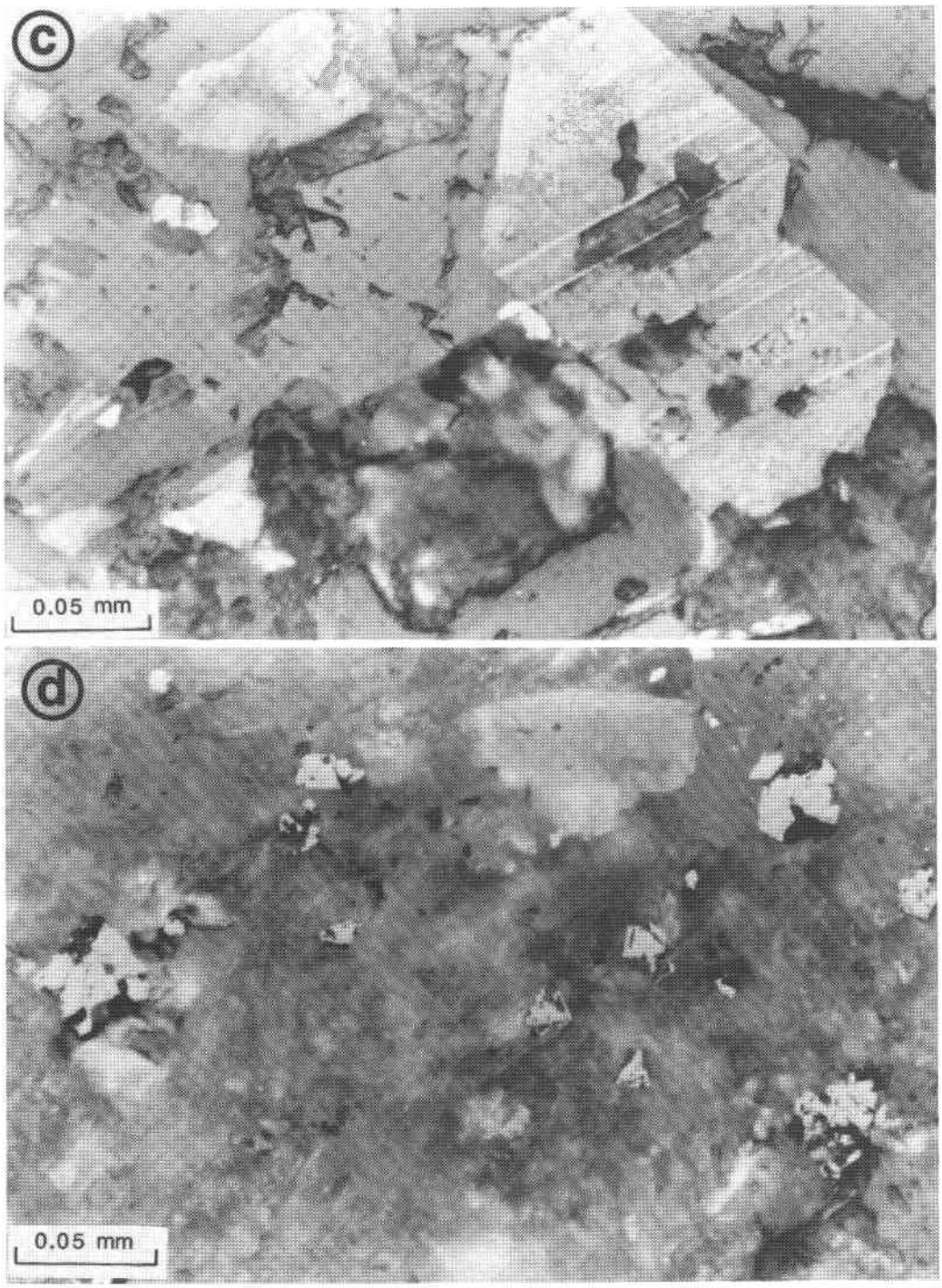


FIG. 5 (*concluded*). (*c*) Rose Harbour swarm, dyke 19, andesite with both normal and reversed magnetization (Figs. 8, 9); large highly altered euhedral grain of titaniferous magnetite with exsolution lamellae of magnetite and ilmenite; highly reflective grains are pyrrhotite. (*d*) Selwyn Inlet swarm, dyke 23, basalt with Bruhnes-age magnetization only (Fig. 10); disseminated grains of uniform unaltered magnetite having no exsolution lamellae.

swarm was not sampled due to time and logistical constraints. Sample locations are shown on Fig. 1.

The dykes range in composition from basalt to rhyolite (Souther and Bakker 1988). They vary from less than 1 to 20 m in thickness, and composite dykes may be as much as 100 m across. Cumulative thickness of dykes is commonly less than 10% of the entire swarm width, but locally they form 50% or more of transects of up to half a kilometre. Generally they are subvertical, sometimes being as much as 10° from the vertical at any one locality. They are only briefly exposed at shorelines, and it is not possible to estimate attitudes over any distance. Inland outcrops are sparse. In any given swarm, up to 60% of the dykes lie within 10° of the mean azimuth for that swarm. Dyke orientation appears to be independent of structures in the wall rock. Most contacts are planar and cut sharply across folds and other structures. Offsets of a few

metres occur locally where dykes are cut by fracture zones. The sense and magnitude of this faulting is widely variable and is probably related to post-emplacement doming during late-stage magmatic or hydrothermal activity.

Age

The dykes have a close spatial association with lithologically similar Tertiary plutonic and volcanic rocks, indicating that they are their coeval and probably cogenetic hypabyssal equivalents. This is borne out by the few K–Ar (hornblende) age dates available (Table 1). The oldest ages (Rose Harbour and Collison Bay, 55, 44, and 41 Ma) are from the south, near the oldest (Eocene) Kano plutons, whereas the youngest (13 Ma) is from the Selwyn Inlet dykes farther north, closer to the younger part of the Masset Formation on Graham Island.

Tectonic implications

Where dyke propagation has been studied in real time, as in Iceland (Sigurdsson 1987), the magma is seen to flow laterally from central magma columns into fractures developed normal to the direction of least compressive stress. Using this relationship, Souther and Jessop (1991) suggested that the Kano plutons occupy the sites of magma columns and that the mean direction of the associated dyke swarms is normal to the direction of least compressive stress during emplacement. Radiometric dates indicate that the predominantly north-trending dykes of the Collison Bay swarm were emplaced during the Middle to Late Eocene, whereas dykes of the east-trending Selwyn Inlet swarm were emplaced during the Middle Miocene (Table 1). It appears, therefore, that east–west extension was followed by north–south extension.

Petrography

Basaltic dykes range from very fine grained, aphyric to microporphyritic assemblages of plagioclase (labradorite–oligoclase), clinopyroxene, opaque oxides, and devitrified glass, to medium-grained ophitic or subophitic dolerites and coarse-grained, hypidiomorphic gabbro. Andesite dykes are commonly medium grained, with microphenocrysts of plagioclase (andesine–oligoclase), augite, and rarely hornblende in a groundmass of the same minerals plus opaque oxides. Dacite and rhyolite dykes commonly contain microphenocrysts of sodic plagioclase and (or) alkali feldspar in a groundmass of alkali feldspar, quartz, opaque minerals, and alteration products. The principal rock types appear to have been emplaced repeatedly during the entire period of intrusion.

The degree of alteration varies from very little, in which the only visible alteration is devitrification of interstitial glass, to intense saussuritization. Alteration is greatest close to subvolcanic plutons, where postintrusion hydrothermal activity has affected both dykes and their wall rocks.

Opaque minerals commonly form less than 5% of dyke rocks. Both oxide and sulfide phases are present as disseminated grains that rarely exceed 0.5 mm in longest dimension. Study of polished sections shows that a wide range of grain sizes, compositions, textures, and oxidation states is present. Titanomagnetite and ilmenite occur both as early-formed euhedral crystals and as anhedral grains occupying irregular voids between silicate minerals. Many of the early-formed titanomagnetite grains have undergone extensive alteration to goethite, maghemite, and rarely hematite. Others now contain exsolution of ilmenite in a magnetite matrix resulting in a very fine, patchy, moss-like or cross-hatched pattern having faint pleochroism. In contrast, the second-generation anhedral grains, particularly of ilmenite, are commonly uniform and show little or no evidence of alteration. The quenched margins of some of the younger dykes contain finely disseminated grains of homogeneous titanomagnetite that contains no visible exsolution lamellae or alteration products.

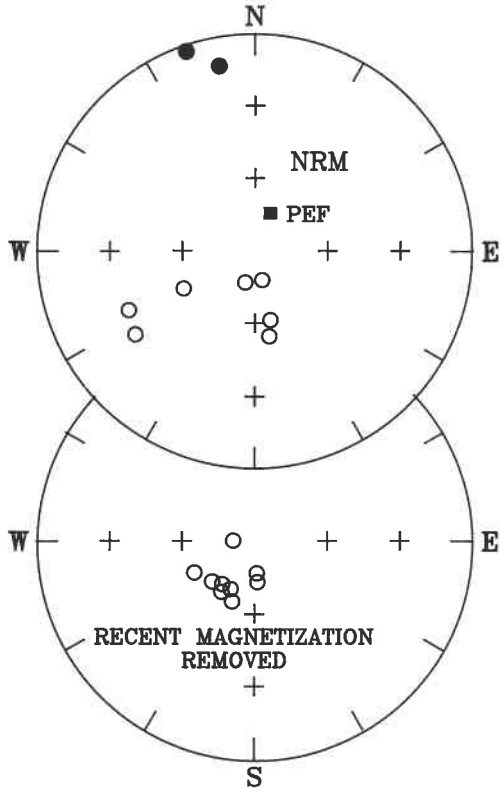


FIG. 6. Directions of magnetization in nine specimens cut from five cores taken from dyke 19 of the Tasu Inlet swarm, an example of reversed magnetization with Brunhes overprint. Above, NRM directions. Below, the directions obtained by line fitting in the range 500–600°C. ●, directions plotted on the lower hemisphere; ○, directions plotted on the upper hemisphere.

Sulphide phases include chalcopyrite and pyrite (both nonmagnetic) and pyrrhotite. Pyrrhotite, which can be a remanence carrier, has a limited distribution, but in a few sections subhedral grains up to 0.5 mm long are the principal opaque mineral.

Sampling

Using an inflatable boat, about 20 representative dykes from each swarm were sampled (Table 2). Altogether 129 coastline sites were occupied (Fig. 2). Five cores were drilled in each dyke and oriented by magnetic compass. The prevailing overcast weather precluded the use of a sun compass, and magnetic bearings were adjusted for any local aberrations by taking sightings to several distant points and applying the appropriate correction to the azimuth of cores. Corrections were rarely more than 2° and commonly less than 1°. Two specimens were cut from each core. From the 678 cores, 1352 specimens were cut and studied.

Paleomagnetism of dyke swarms

Techniques

Magnetic measurements were made on automated Schonstedt spinner magnetometers, which allow batches of 15 specimens to be measured unattended. Stepwise thermal demagnetization

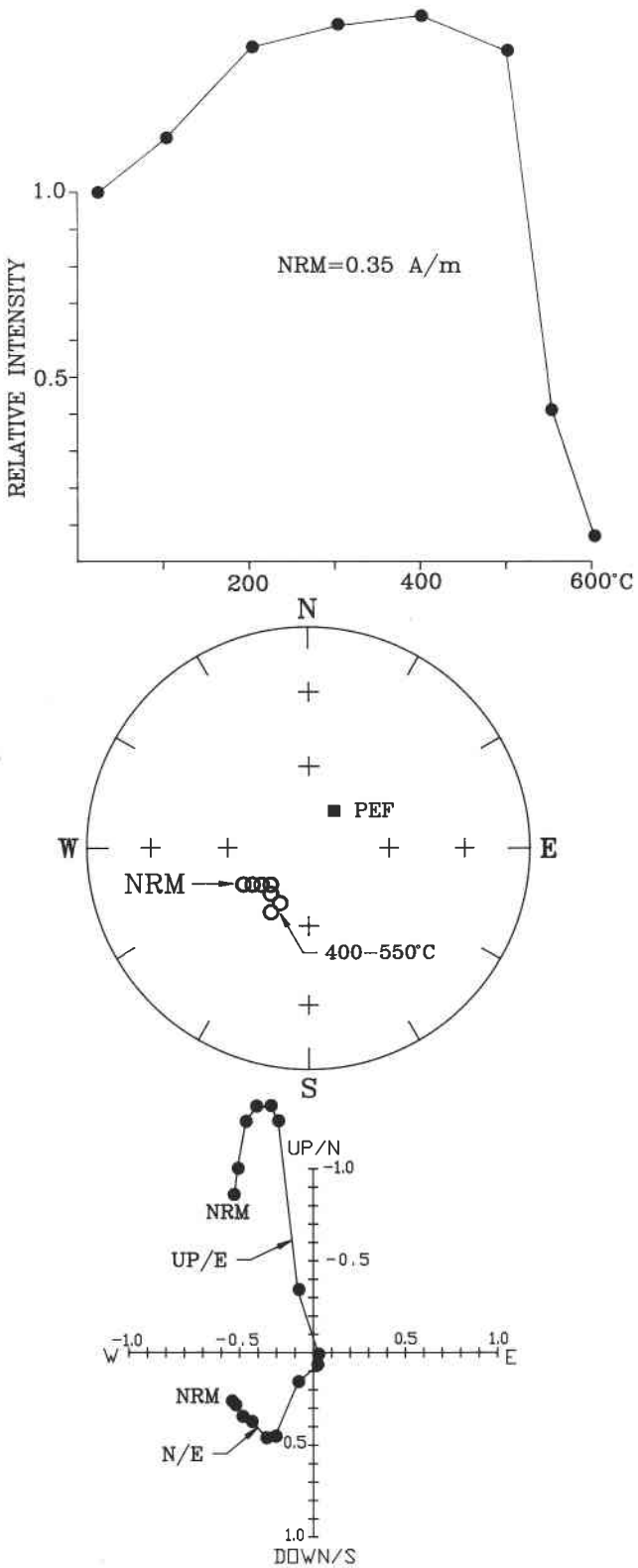


FIG. 7. Thermal demagnetization of specimen from dyke 19 of the Tasu Inlet swarm. Above, the changes of intensity with temperature. Middle the corresponding changes in direction. Below, normalized orthogonal plots. See Fig. 6 for all specimen directions from this dyke.

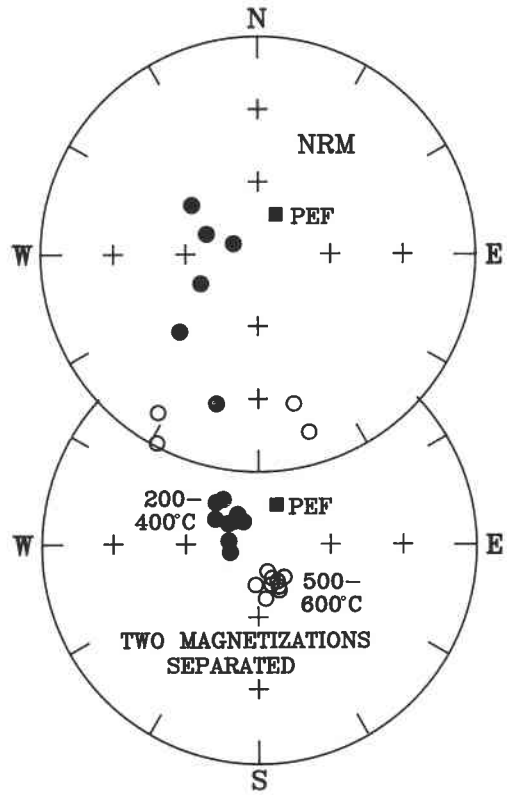


FIG. 8. Direction of magnetization in 10 specimens cut from five cores taken from dyke 19 of the Rose Harbour swarm, an example of low-T_{UB} normal and high-T_{UB} reversed magnetizations occurring in the same specimen. Above, NRM directions. Below, the separated normal (●) and reversed (○) magnetizations.

was carried out using Schonstedt TSD1 demagnetizers. Alternating field (AF) demagnetization up to 100 mT (peak field) was carried out using a Schonstedt GSD-5 tumbling demagnetizer. For higher fields, up to 175 mT, a Sapphire SI4 instrument was used, demagnetizations being made along three axes sequentially. After measurement of the natural remanent magnetization (NRM), two pilot specimens from each dyke were chosen (258 altogether), one for AF and the other for thermal demagnetization. They were demagnetized in 8–10 steps. Magnetizations were identified as stable end points (McElhinny and Gough 1963) or as lines on orthogonal plots (Zijderveld 1968; Kirschvink 1980).

To determine whether the range of demagnetization within which lines were observed in pilot specimens from a given dyke was typical of all specimens from that dyke, three to five further specimens were chosen and demagnetized in the same range, usually in three or four steps. If decay was linear to the origin, then, within the linear range, a field strength or temperature was chosen at which all other specimens from that dyke were treated. Often the ranges were not the same, in which case all remaining specimens from the dyke were demagnetized at about five steps (six or more if both normal and reversed magnetizations are present) and linear ranges determined individually for each. Five main types of magnetizations were recognized.

Dykes with single stable magnetizations

Most pilot specimens showed simple behaviour. After treatment in low AF or at low temperatures their directions stabilized, as magnetizations variable in magnitude and directed either close to the present Earth's magnetic field (PEF) or to the geocentric axial dipole field (GAD) were removed. These latter magnetizations are probably viscous remanent magnetizations (VRM) acquired since the last reversal of the field and

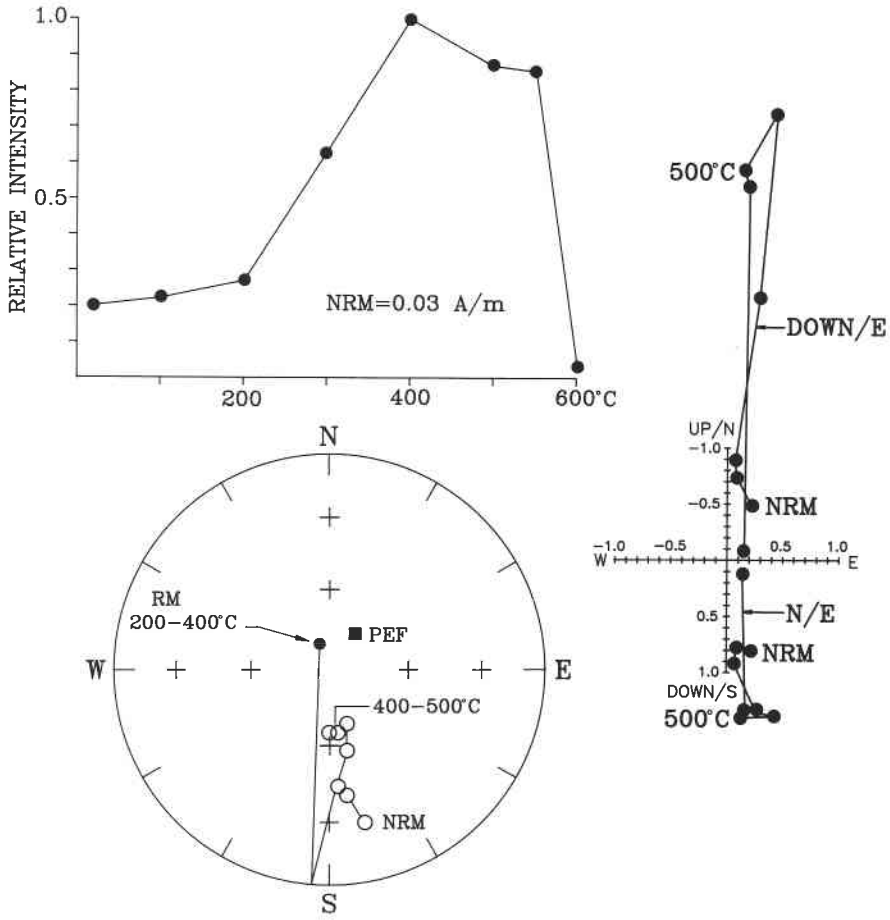


FIG. 9. Thermal demagnetization of specimen from dyke 19 of the Rose Harbour swarm. The three diagrams are the same as in Fig. 7. See Fig. 8 for all specimen directions from this dyke.

are referred to as Brunhes overprints. At higher demagnetizing fields and temperatures, such specimens have stable end points and orthogonal plots were linear and converged on the origin. The magnetizations are of either normal or reversed polarity. Altogether 89 dykes showed this comparatively simple behaviour (Table 2).

In the example of Fig. 3 the NRM directions are vertically downwards (normal polarity) and are very well grouped. Directions and intensities remain virtually unchanged up to 550°C (Fig. 4). Only one magnetization is present with unblocking temperatures (T_{UB}'s) between 550 and 600°C. Study of a polished section shows that the principal opaque mineral is ilmenite, which is present as relatively large (0.1 mm) grains and contains patches of extremely small exsolved granules. These are too small to identify but are believed to be magnetite (Fig. 5*a*). Presumably it is these very small granules that carry the stable normal magnetization.

In other dykes the NRM directions are scattered (Fig. 6). The intensity increases initially during thermal demagnetization as a Brunhes overprint is removed and an underlying reversed magnetization is removed (Fig. 7). Above 500°C, the orthogonal plots decay linearly to the origin, and the directions from different specimens become well grouped. The direction of the stable remanence can be accurately estimated by line fitting in the range 500–600°C, which is the range of T_{UB}'s present. The rock contains relatively large pseudomorphs of titanomagnetite (Fig. 5*b*). Exsolution and alteration have resulted in numerous small isolated magnetite grains, which are probably responsible for the stable magnetization. Figures 5*a* and 5*b* would correspond to class III or higher oxidation state of Wilson and Watkins (1967).

Dykes with both normal and reversed magnetization

Orthogonal plots from many pilot specimens revealed two magnetizations, one normal and the other reversed, each with a distinctive T_{UB} range, respectively 200–400 and 500–600°C (Figs. 8, 9). In the example the direction migrates along a great circle to a stable end point between 400 and 550°C. The NRM direction and that measured after treatment at 100°C do not fall on this great circle, but on one through the PEF (Fig. 9). Hence a small Brunhes overprint also is present and is removed by heating to 200°C. The magnetizations removed at higher temperatures (RM of Fig. 9) are not Brunhes overprints because their directions are well grouped and significantly different in direction from the PEF or the GAD (Fig. 8). Polished sections show complex altered titanomagnetites with exsolved magnetite granules, and disseminated pyrrhotite is also present (Fig. 5*c*). The latter probably carries the low-T_{UB} normal magnetization and the former carry the high-T_{UB} reversed magnetization. This is unlikely to reflect a reversal of the field that occurred during the time taken for the dyke to cool. Nor is it likely to record a self-reversal of magnetization (Uyeda 1958) because the directions in individual specimens are not exactly opposed to one another (Figs. 8, 9). More probably, one magnetization was acquired upon intrusion, whereas the other was acquired during secondary hydro-

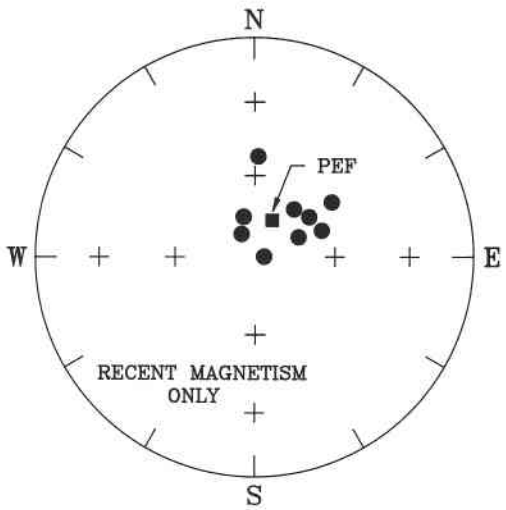


FIG. 10. Directions of magnetization in nine specimens taken from five cores from dyke 23 of the Selwyn Inlet swarm, an example of a dyke containing Brunhes magnetization only.

thermal alteration long afterwards, when the field was of opposite polarity. It is not clear which is the older. A total of 12 dykes showed this complex behaviour (Table 2).

Dykes with Brunhes magnetizations only

Fourteen dykes had directions of NRM that were clustered loosely about the direction of the recent field (Fig. 10). Intensities fell rapidly during both AF and thermal demagnetization and directions became widely scattered (Figs. 11, 12). The magnetizations appear to be entirely Brunhes overprints. The opaque minerals (Fig. 5*d*) comprise small grains of ilmenite, magnetite, and (or) titaniferous magnetite, all of which are relatively homogeneous and unaltered; they correspond to class I oxidation state of Wilson and Watkins (1967). The absence of exsolution textures and the fine grain size of the rock indicate rapid cooling and little postemplacement alteration. The absence of ultrafine exsolution textures would account for the low stability and recent age of the magnetization.

Dykes with incoherent magnetization

Twelve dykes have widely scattered directions that demagnetization does not resolve. Orthogonal plots contain no linear segments (Fig. 13). Sulfides are abundant and the iron–titanium oxides have been extensively replaced by goethite and (or) hematite. Silicate minerals also are intensely altered, probably by late-stage hydrothermal activity.

Dykes affected by lightning

Two dykes have very strong magnetizations (10^2 A/m) and scattered directions. During AF demagnetization, their directions either did not change or migrated along great circles without reaching end points. These dykes have probably been hit by lightning.

Of the 129 dykes studied, 101 yield identifiable magnetizations that can be regarded as records of the paleofield and 28 provide no such information. The types of magnetization present in each swarm are summarized in Tables 2 and 3. Eighty nine dykes have single magnetizations, 32 normal, and 57 reversed. The remainder (12) contain both normal and reversed magnetizations. Evidently most dyke records represent the paleofield over a short interval of time, but others span a reversal of the field. Statistics for each dyke and for each

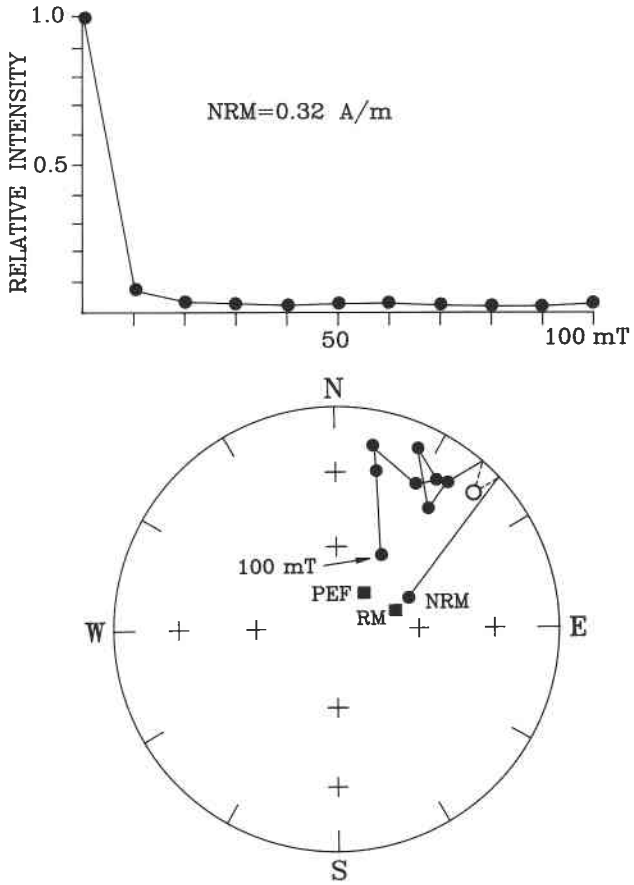


FIG. 11. Alternating-field demagnetization of specimen from dyke 23 of the Selwyn Inlet swarm. Above, intensity. Below, direction changes. RM is the direction of magnetization obtained by line fitting in the range 0–20 mT. All specimen directions from this dyke are shown in Fig. 10.

swarm are given in Tables 3 and 4. The mean directions of magnetization observed from dykes with normal and (or) reversed magnetization are plotted in Fig. 14.

Paleomagnetism of lavas of Masset Inlet

In the early study of Hicken and Irving (1977) samples were collected from 13 sites around Masset Inlet and 2 sites on Ian Lake (Table 3). Lavas in the vicinity have yielded K–Ar dates ranging from 25 to 19 Ma, or Early Miocene (Hickson 1991). Each site was an individual flow, except site 10, which was a dyke intruding volcanic breccia. Bedding features are commonly lacking, and estimation of bedding at the sampling sites is not possible. Elsewhere in the Masset Inlet region, dips of 10–15° have been recorded (Sutherland Brown 1968; Hickson 1991), but vent positions are unknown so it is not possible to determine if this is a paleoslope or later tilting. Four flows yielded incoherent magnetization. Excellent data were obtained from eight sites. Four have normal polarity and four are reversed.

Paleomagnetism of Louise Island pluton

In the same earlier study, samples were collected from five sites in massive quartz monzonite on the eastern shore of Carmichael Passage. A U–Pb zircon age of 34.3 ± 0.6 Ma has been obtained by Anderson and Reichenbach (1991). Brunhes overprints often comprise 80% of the NRM, but stable end points and linear trajectories occur in the range 10–100 mT.

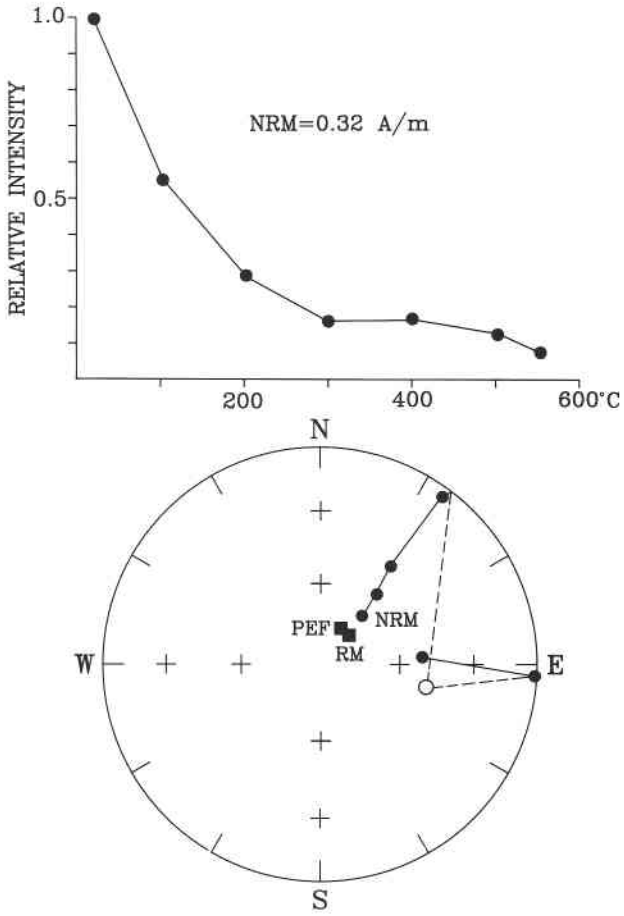


FIG. 12. Thermal demagnetization of specimen from dyke 23 of the Selwyn Inlet swarm. Above, intensity. Below, direction changes. RM is the direction of magnetization obtained by line fitting in the range 20–300°C.

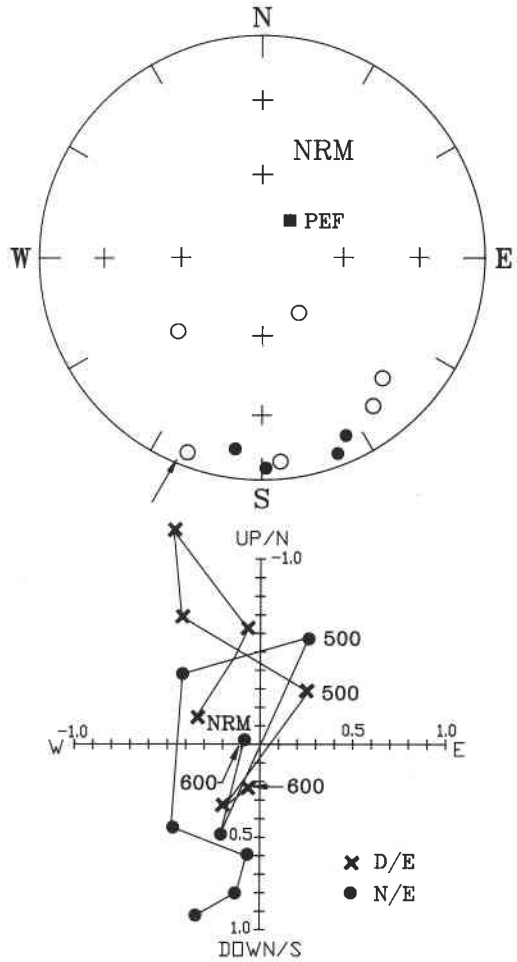


FIG. 13. Above, directions of magnetization in 10 specimens cut from six cores from dyke 18 of the Burnaby Island swarm, an example of scattered magnetizations that demagnetization does not resolve. Below, orthogonal plots of thermal demagnetization data (in °C) are irregular and lines are not present; the arrow marks the NRM direction of this specimen.

T_{UB}'s range as low as 100°C (corresponding to Brunhes overprints), but small high T_{UB}'s are present, and there are excellent end points between 400 and 600°C. Normal and reversed polarities are present, and site directions are very well grouped (Fig. 14, Table 4). AF demagnetization yielded excellent lines between 10 and 100 mT.

Summary of paleomagnetic data

Mean dyke directions for each swarm are plotted in Fig. 14 along with mean direction at sampling site in the Masset Inlet lavas and Louise Island pluton. Mean directions of magnetization for each swarm and for lavas and pluton are given in Table 4. Means for normal and reversed collecting localities are listed separately, and then together irrespective of sign. In these summary calculations each site is given unit weight. Sites with errors (α_{95}) exceeding 15° are not included. Paleopoles for each swarm are given as means of virtual geomagnetic poles (VGP's) obtained from collecting sites. The most notable feature of Fig. 14 and Table 4 is the steepness of the mean inclinations (about 80°), which are appropriate for a latitude of about 73°, about 15° higher than at present. This is true for all dyke swarms, Masset Inlet lavas, and the Louise Island pluton.

Comparison with mainland North America

Middle Eocene to Middle Miocene paleopoles are available from mainland North America outside regions of recent tectonic disturbance (Table 5, Fig. 15). This broad interval (50–10 Ma) has been chosen so as to span generously the time represented by the rocks studied herein. Paleopoles have been used only if they are based on data from 10 or more collecting sites, and have individual errors (A_{95}) less than 15°. Their grouping is good, and their overall mean is defined to an accuracy of better than 3° (Table 5, entry 14). In Table 5 and Fig. 16 averages for three time intervals each of 20 Ma duration (50–30, 40–20, and 30–10 Ma) are given. Relative to North America, the paleopole seems to have moved very little if at all for a period of about 40 Ma, so comparisons can be made with our data with negligible uncertainty as to time correspondences.

On the basis of their radiometric ages, the rock units studied herein have been assigned to one of three age ranges. Assuming that they have not moved relative to mainland North America, the geomagnetic field direction expected for each rock unit has been calculated (Table 6), using the mean paleopoles of Table 5, entries 11–13. The expected directions, using the full range (50–10 Ma, entry 14 of Table 5) represented by mainland paleopoles, have also been calculated. Expected directions always have much lower inclinations than those observed. Departures in inclination are independent of swarm age and swarm trend. In the two youngest units (Selwyn Inlet swarm,

TABLE 3. Data by sites

Site (polarity)	UTM (E,N)	M (A/m)	n	s	D (°)	I (°)	k_S	α_{95} (°)	Treatment
					Selwyn Inlet swarm				
01(N)	305800, 5863900	3.395	5	10	73	79	377	3	550°C(1), 15 mT(9)
02(N)	306150, 5863100	0.866	5	10	13	76	38	8	100–400°C(1), 10 mT(9)
03(R)	308150, 5861350	1.816	5	10	132	−86	168	4	500–600°C(1), 20 mT(9)
04(R)	307800, 5860100	0.812	5	10	249	−82	975	2	500–600°C(1), 20 mT(9)
05(R)	307700, 5860000	0.600	5	10	282	−83	270	3	500°C(1), 20 mT(9)
09(N)	309350, 5862550	0.784	6	12	9	77	12	13	500–700°C(1), 10 mT(11)
10(R)	309700, 5862200	0.212	5	8	222	−78	37	9	500–600°C(1), 20–30 mT(7)
11(N)	311200, 5861450	1.344	5	9	349	72	133	5	100–300°C(1), 10–70 mT(8)
12(N)	311200, 5861450	1.111	6	9	0	72	259	3	100–400°C(1), 10–70 mT(8)
13(N)	309050, 5863050	0.924	5	10	338	71	132	4	100–600°C(1), 10–50 mT(5), 10–30 mT(4)
14(N)	316700, 5858800	0.441	5	9	96	72	30	10	100–300°C(9)
14(R)	316700, 5858800	—	5	10	302	−49	40	8	450–600°C(9), 10–30 mT(1)
15(N)	316700, 5858800	0.700	5	9	332	75	31	9	100–300°C(9)
15(R)	316700, 5858800	0.644	2	2	259	−56	48	37[a]	500–600°C(2)
16(R)	316750, 5857900	0.307	5	10	285	−70	381	3	400–600°C(1), 20–60 mT(9)
17(R)	316800, 5858150	1.136	7	14	271	−65	417	2	300–600°C(1), 20–60 mT(13)
18(N)	313550, 5859600	0.985	5	10	12	70	161	4	100–300°C(9), 10–60 mT(1)
18(R)	313550, 5859600	—	5	9	250	−69	165	4	450–600°C(9)
19(N)	313100, 5859250	0.270	5	5	113	82	15	21[a]	150–300°C(5)
19(R)	313100, 5859250	—	5	10	209	−70	138	4	400–550°C(5), 10–50 mT(5)
21(N)	312350, 5861900	0.070	5	9	317	73	208	4	30–70 mT(9)
22(N)	312750, 5859000	0.684	6	12	350	69	319	2	100–300°C(11), 20–80 mT(1)
22(R)	312750, 5859000	—	6	11	248	−67	270	3	450–600°C(11)
24(N)	313250, 5859350	0.887	5	10	347	76	242	3	100–300°C(1), 20–80 mT(9)
24(R)	313250, 5859350	—	5	10	189	−78	22	11	500–600°C(1), 125–175 mT(9)
					Tasu Sound swarm				
02(R)	300650, 5853800	0.013	5	10	168	−74	159	4	500°C(9), 30 mT(1)
03(R)	298100, 5852850	0.009	6	11	151	−82	172	4	500°C(11)
04(N)	298450, 5853200	1.061	5	10	323	87	87	5	500–600°C(9), 20–50 mT(1)
07(R)	301300, 5849600	0.340	5	8	177	−81	67	7	200–400°(6), 20–60 mT(2)
08(N)	300550, 5849200	0.022	5	7	291	81	35	10	550°C(3), 80 mT(2)
08(R)	300550, 5849200	—	6	9	178	−69	25	11	400–450°C(2), 400–550°C(3), 40–60 mT(4)
10(N)	300600, 5849100	0.590	5	10	48	85	177	3	550°C(3), 20 mT(7)
11(N)	301750, 5846650	3.816	5	10	297	88	1565	1	500°C(1), 20 mT(9)
12(N)	302500, 5845500	7.704	5	10	249	78	1226	1	500°C(1), 20 mT(9)
13(R)	298200, 5854650	1.306	5	9	168	−76	513	2	500°C(4), 30 mT(5)
14(R)	298100, 5853150	0.265	5	10	175	−72	218	3	550°C(2), 20 mT(8)
15(N)	297450, 5852450	0.008	5	8	319	73	47	8	100–250°C(7), 10–70 mT(1)
15(R)	297450, 5852450	—	4	7	213	−65	12	18[a]	300–500°C(2), 450–500°C(5)
16(N)	297700, 5852450	0.003	5	8	1	73	53	8	550–600°C(5), 50–80 mT(1), 80–150 mT(2)
16(R)	297900, 5852450	—	5	10	165	−69	36	10	400–500°C(5), 10–30 mT(1), 20–60 mT(4)
18(R)	298150, 5855300	0.308	5	10	128	−62	155	4	500–600°C(9), 20 mT(1)
19(R)	298100, 5855700	0.528	5	9	213	−74	70	6	500–550°C(5), 500–600°(3), 20 mT(1)
					Lyell Island swarm (including Bigsby Inlet swarm)				
01(R)	325100, 5834900	1.419	7	13	169	−68	154	3	550°C(1), 20 mT(12)
02(N)	324100, 5835500	2.097	5	10	302	69	47	7	300°(1), 5 mT(6), 10 mT(3)
03(R)	323650, 5835400	1.428	6	13	219	−69	30	8	500–600°C(1), 10–40 mT(12)
04(R)	324000, 5834050	2.032	5	11	176	−78	926	2	550°C(1), 20 mT(10)
07(N)	331900, 5832800	0.211	5	11	357	80	64	6	100–500°C(1), 30 mT(10)
08(N)	331900, 5832800	0.076	5	9	77	70	39	8	500–600°C(1), 10–40 mT(8)
09(R)	332700, 5832000	0.005	5	7	112	−78	20	14	500°(6), 20–40 mT(1)
10(R)	325500, 5833350	1.361	5	9	241	−78	1175	2	550°(1), 10 mT(8)
11(N)	325650, 5833300	0.434	5	10	31	76	583	2	550°C(9), 10–20 mT(1)
12(R)	326800, 5832100	0.003	5	9	164	−76	262	3	550°C(1), 10 mT(9)
16(R)	327550, 5830150	0.001	5	10	31	−80	12	15	400°C(1), 20 mT(6), 30 mT(3)
17(R)	319100, 5838650	0.993	5	10	166	−71	1830	1	500°C(1), 20 mT(9)
18(N)	319000, 5838700	1.801	6	12	332	69	1072	1	500°(1), 20 mT(11)
19(R)	325700, 5842900	0.231	5	10	182	−79	50	7	550–650°C(2), 20 mT(4), 30 mT(4)
20(R)	312750, 5842200	2.593	5	10	107	−84	1375	1	500–550°C(1), 20 mT(9)
21(N)	316300, 5833800	0.008	5	10	344	84	94	5	300°C(1), 10 mT(9)
24(N)	317100, 5833400	0.006	5	10	5	87	311	3	500°C(1), 10 mT(9)
25(N)	313150, 5834800	2.848	6	12	317	84	554	2	500°C(1), 20 mT(11)

TABLE 3 (*continued*)

Site (polarity)	UTM (E,N)	M (A/m)	n	s	D (°)	I (°)	k_S	α_{95} (°)	Treatment
					Burnaby Island swarm				
01(N)	347100, 5809850	0.441	5	10	4	74	177	4	100–300°C(1), 10 mT(9)
02(N)	346850, 5809700	1.215	4	5	315	60	8	30[a]	550°C(4), 10 mT(1)
02(R)	346850, 5809700	0.263	2	3	234	−80	7	50[a]	5–40 mT(1), 10–20 mT(2)
03(R)	346050, 5809100	0.217	6	12	248	−83	382	2	300–500°C(1), 20 mT(1)
04(R)	346700, 5807050	0.517	3	5	159	−76	69	9	20–30 mT(3), 20–60 mT(2)
05(N)	346600, 5807000	0.193	3	6	26	64	27	13	100–300°C(1), 10–60 mT(5)
05(R)	346600, 5807000	—	5	10	232	−84	289	3	300–550°C(1), 60 mT(9)
06(R)	345350, 5807900	0.368	6	12	134	−63	15	12	500–600°C(1), 20 mT(11)
07(R)	345250, 5804800	0.804	5	10	195	−83	30	9	400–600°C(1), 15 mT(9)
08(R)	346500, 5804550	0.085	6	12	171	−41	78	5	300–600°C(1), 30 mT(11)
09(R)	341950, 5813000	0.966	5	10	305	−84	74	6	400–600°C(1), 20 mT(9)
10(R)	342850, 5812700	0.854	5	10	83	−68	99	5	400–600°C(1), 30 mT(9)
11(R)	342000, 5811900	0.014	5	10	149	−85	346	3	400°C(3), 450°C(6), 50 mT(1)
12(R)	339450, 5813000	0.103	4	8	168	−77	28	11	200–400°C(6), 10–20 mT(1), 20 mT(1)
13(R)	339400, 5813800	0.037	6	12	80	−73	17	11	200–300°C(1), 5 mT(9), 10 mT(2)
15(R)	339500, 5815350	0.007	5	10	240	−81	11	15	300–500°C(1), 50 mT(9)
16(R)	339150, 5815200	0.017	5	10	310	−86	111	5	400°C(9), 60 mT(1)
19(R)	339750, 5811650	0.806	5	10	90	−70	385	3	500–600°C(8), 20–40 mT(2)
20(R)	344300, 5801750	0.003	5	10	187	−74	55	7	300–500°C(3), 40 mT(7)
					Collison Bay swarm				
02(N)	344900, 5796350	3.136	6	12	19	82	568	2	500°C(1), 20 mT(11)
03(N)	345250, 5796100	0.297	5	10	261	88	662	2	500°C(1), 20 mT(9)
04(R)	349850, 5797550	0.552	5	10	242	−86	106	5	200–400°C(9), 10–30 mT(1)
05(R)	350550, 5798400	0.003	5	10	187	−79	142	4	400–600°C(5), 500–600°C(4), 40–80 mT(1)
06(R)	350850, 5798600	0.004	5	10	176	−73	64	6	500°(9), 20–70 mT(1)
07(N)	351350, 5798750	0.011	5	10	321	61	34	8	200°C(2), 30 mT(8)
08(R)	351350, 5798750	0.003	5	10	197	−82	210	3	500°C(9), 20–60 mT(1)
09(R)	351600, 5799050	0.009	5	10	180	−73	78	6	500°C(9), 90 mT(1)
10(R)	351800, 5799200	0.013	5	10	169	−79	697	2	500°C(9), 20–80 mT(1)
11(N)	352000, 5799350	0.485	5	10	18	85	31	9	300–600°C(1), 10 mT(9)
12(N)	359950, 5798700	0.157	5	10	315	78	35	8	200°C(4), 300°C(5), 10–20 mT(1)
13(N)	359950, 5798700	0.020	5	10	314	74	185	4	100–400°C(1), 20 mT(9)
14(N)	353700, 5794250	0.058	5	10	338	69	81	5	500°C(1), 20 mT(7), 10–50 mT(2)
16(N)	355150, 5794450	0.152	5	10	335	74	64	6	100–400°C(1), 20 mT(9)
17(N)	345250, 5795500	1.985	5	10	351	88	1377	1	500°C(1), 20 mT(9)
18(R)	350050, 5801750	0.509	5	10	161	−77	126	4	500°C(3), 20 mT(7)
19(R)	346650, 5796750	0.004	5	9	169	−65	285	3	300°C(7), 400°(2)
20(N)	344000, 5800550	0.082	5	10	11	68	531	2	500°C(1), 20 mT(9)
					Rose Harbour swarm				
02(N)	357450, 5787850	0.557	5	9	188	75	98	5	300–500°C(1), 10–50 mT(8)
03(N)	357450, 5787850	0.012	6	12	182	71	22	9	300–600°C(1), 20–80 mT(11)
04(R)	357050, 5788000	0.217	8	16	163	−70	84	4	300°C(1), 5 mT(15)
05(R)	356950, 5787950	0.004	5	10	140	−79	49	7	300–500°C(1), 20–60 mT(9)
07(R)	356700, 5787950	0.005	6	12	179	−77	122	4	500°C(11), 30 mT(1)
08(R)	356650, 5787800	0.455	5	10	123	−65	262	3	500°C(7), 20–50 mT(3)
10(N)	353900, 5788950	0.015	5	10	287	78	387	3	500°C(6), 30 mT(4)
11(N)	355050, 5787100	1.246	5	10	311	72	343	3	200–400°C(1), 20 mT(9)
12(R)	352100, 5788300	0.150	5	10	79	−88	1456	1	500°C(9), 20 mT(1)
13(R)	360300, 5780950	0.231	5	10	160	−81	289	3	300–650°C(1), 20–60 mT(9)
16(R)	358750, 5781000	0.015	6	12	27	−83	133	4	300–550°C(3), 60 mT(9)
17(R)	353600, 5780900	0.503	5	10	137	−81	253	3	400–600°C(3), 10–50 mT(7)
18(R)	353650, 5782250	0.004	5	10	196	−57	64	6	500°C(7), 550°C(2), 30 mT(1)
19(N)	355050, 5781500	0.031	5	9	310	75	123	5	200–400°C(9)
19(R)	355050, 5781500	0.031	5	9	163	−75	326	3	500–625°C(9)
20(R)	355100, 5782500	0.003	5	10	172	−70	88	5	500°C(3), 300–600°C(3), 20–60 mT(3)
					Volcanics of Masset Inlet				
04(R)	668500, 5949200	1.348	7	15	242	−76	154	3	550°C(1), 30 mT(14)
05(R)	668600, 5941500	0.290	8	14	222	−78	33	7	25 mT(14)
06(N)	663200, 5949800	0.968	6	11	71	76	1081	1	40 mT(11)
07(N)	663300, 5949800	1.606	7	15	46	72	94	4	400°C(1), 40 mT(14)

TABLE 3 (*concluded*)

Site (polarity)	UTM (E,N)	M (A/m)	n	s	D (°)	I (°)	k_S	α_{95} (°)	Treatment
08(R)	667250, 5934040	19.033	8	16	139	−83	383	2	550°C(1), 40 mT(15)
09(R)	667250, 5940500	11.597	6	12	169	−81	308	3	40 mT(12)
10(N)	668100, 5941900	4.771	5	9	58	82	54	7	400°C(1), 25 mT(8)
11(N)	668100, 5942000	1.626	5	9	322	88	55	7	25 mT(9)
					Louise Island pluton				
08(R)	303700, 5876000	0.309	6	11	181	−81	163	4	25 mT(11)
09(R)	303800, 5876300	0.197	6	11	133	−83	197	3	550°C(1), 25 mT(10)
10(R)	303600, 5875900	0.390	5	10	146	−76	19	11	25 mT(10)
11(N)	304400, 5873800	0.202	5	10	338	77	36	8	25 mT(10)
13(N)	304300, 5873800	0.268	5	10	350	81	46	7	400°C(1), 25 mT(9)

NOTES: Data are from 101 dykes from which coherent magnetizations were obtained. N and R in column one refer to normal and reversed magnetizations sometimes occurring in the same dyke. M, intensity of magnetization; n, the number of cores; s, number of specimens cut from the n cores; D and I, declination and inclination of mean direction of magnetizations of s specimens given with respect to present horizontal; k_s, Fisher's precision; and α_{95}, radius of the circle of the confidence at $P = 0.05$. In the last column the procedures by which the magnetization was obtained are listed. A single treatment value (e.g., 15 mT or 550°C) signifies single cleaning at that level reflecting a stable end-point direction. When a range is quoted (e.g., 100−400°C) the magnetization has been obtained by least-squares fitting to decay lines observed in that range. The number of specimen treated is given in parentheses. Data from Masset Inlet and Louise Island were summarized previously (Hicken and Irving 1977) and are set out here in detail. New data obtained by thermal demagnetization are included. In the original paper, data from sites 8 to 11 from Masset Inlet were corrected for small northerly bedding tilts of 5−10° recorded from nearby outcrops on the map of Sutherland Brown (1968). Here they are given relative to present horizontal. Site 10 is from Masset Inlet dyke.

[a]Value from this site rejected in general analysis of Table 5 because $\alpha_{95} > 15°$.

and Masset Inlet lavas) declinations also are anomalous, being more easterly than expected. The anomalous nature of the Queen Charlotte Island data is expressed in terms of paleopoles in Fig. 16. All are near sided. Departures from North American reference paleopoles are individually and collectively significant. They cannot have arisen from inadequate time averaging of the geomagnetic field, because in each study reversals are present, and collectively the rock units span 30 Ma or more.

Causes of near-sided paleopoles

Near-sidedness cannot be explained by rotation about vertical axes. No possible rotation will bring near-sided paleopoles into agreement with those from mainland North America (Fig. 16), a result which does not support the rotations invoked by Hicken and Irving (1977) and Higgs (1990, 1991). The former authors based their speculations on paleomagnetic data from lavas of Masset Inlet (Fig. 1, MI of Fig. 16), which they had assumed to be of Paleocene age. Based on the only age then available (K−Ar 62 Ma; Sutherland Brown 1968, pp. 105, 118), it was not then known that the Tertiary lavas of the Queen Charlotte Islands were erupted over an extended interval of 65−10 Ma. The assignment of Masset Inlet volcanics to the Miocene (Hickson 1991) invalidates the interpretation of Hicken and Irving (1977).

Inaccurate recording of the field could be a cause of near-sided poles (or oversteepened inclinations), because of magnetic anisotropy for example. To test this, one specimen from each swarm was selected and a steady field of 100 mT applied to it. The isothermal remanent magnetization (IRM) produced was then demagnetized. The experiment was repeated, applying the field successively in orthogonal directions. The results from six specimens are as follows: IRM_X $D = 000°$, $I = 01°$, $k = 622$, $\alpha_{95} = 3°$; IRM_X (after AF of 10 mT) $D = 000°$, $I = 01°$, $k = 636$, $\alpha_{95} = 3°$; IRM_Y $D = 091°$, $I = 00°$, $k = 1010$, $\alpha_{95} = 2°$; IRM_Y (10 mT) $D = 091°$, $I = 00°$, $k = 358$, $\alpha_{95} = 4°$; IRM_Z $D = 127°$, $I = 89°$, $k = 969$, $\alpha_{95} = 2°$; IRM_Z (10 mT) $D = 137°$, $I = 89°$, $k = 669$, $\alpha_{95} = 3°$. There are no significant departures of directions from the applied field, indicating that no significant remanence anisotropy is present. A further possibility is that the magnetization could have been deflected towards the planes of dykes under various special circumstances described by Strangway (1961). But oversteepening is shared by the pluton and lavas, bodies with quite different shapes. Moreover, in older dykes with approximately north−south trends, such deflections would not cause oversteepening because the field already was close to their planes.

Near-sidedness might reflect latitudinal motion, in this instance from the north by about 20°. The Queen Charlotte Islands would have been in the latitude of northern Alaska during the middle Tertiary, a displacement that is very unlikely because it is in the opposite sense to that anticipated from Pacific plate motions and because paleomagnetic data from Eocene rocks of Vancouver Island and from mainland British Columbia are concordant (Bardoux and Irving 1989; Symons and Welling 1989; Irving and Brandon 1990; Vandall and Palmer 1990). Indeed, if latitudinal motion has occurred, it is likely to have been of the order of tens of kilometres and at most 100 km from the south and to have been associated with the Tertiary opening of the Queen Charlotte Basin. Such small motions are not detectable paleomagnetically and are not evident in our observations.

There remains the possibility that the near-sidedness of Queen Charlotte Island paleopoles was caused by tilting. The tilts about horizontal axes required to bring the mean directions of each rock unit into agreement with the expected fields are listed in Table 6 and plotted in Fig. 17. Apparent tilts are downwards to the northwest, essentially parallel to the present Queen Charlotte Island Fault for the younger Selwyn Inlet swarm and Masset Inlet volcanics and generally towards the north in older rock units. The tilts range between 9° and 15°.

Alternatively, the apparent tilts could have a composite origin. For example, the apparent north-northwesterly tilts could have been produced by a northerly tilt followed by an anticlockwise

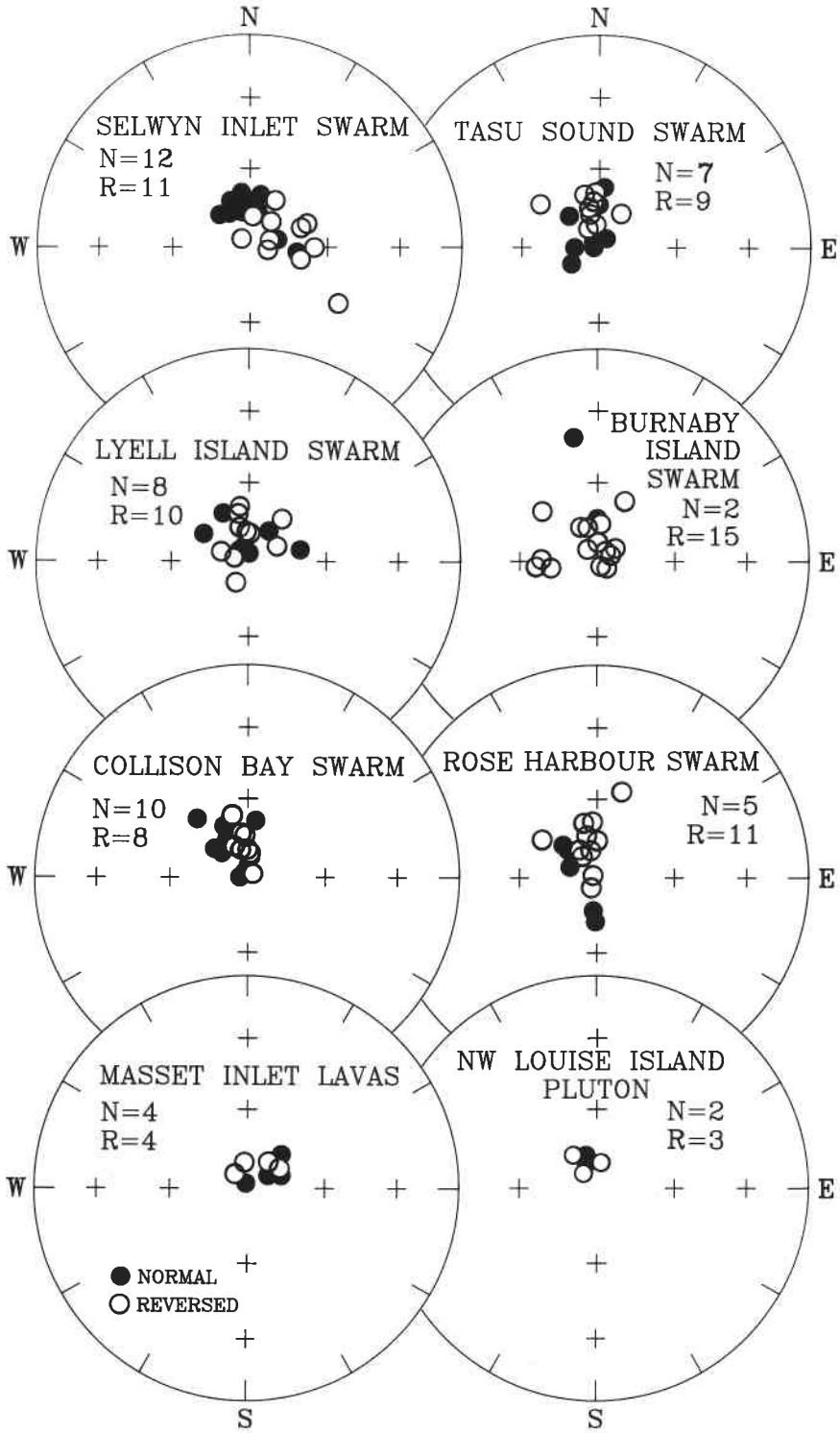


FIG. 14. Mean directions of magnetization at collecting sites in dykes, lavas, and pluton. ●, north-seeking direction (normal polarity) plotted on the lower hemisphere; ○, south-seeking directions (reversed polarity inverted through the origin) also on the lower hemisphere, allowing normal and reversed directions to be compared directly. Data from sites with Brunhes magnetization, with incoherent magnetizations, and that have been struck by lightning are not plotted. The perimeter is always the present horizontal plane.

rotation about vertical axes. Although there is no evidence against such complex explanations, which may need to be considered eventually, it is assumed as a first approximation that the near-sidedness of Queen Charlotte Island paleopoles is the product of single tilts (Fig. 17).

Proposed stress history

The presence of the dyke swarms and the occurrence of tilts inferred from their paleomagnetism indicate that the south and centre of the Queen Charlotte Islands have undergone wide-

TABLE 4. Summary of data

Rock unit	B	D_m (°)	I_m (°)	k_m	α_{95} (°)	λN (°)	φE (°)
SI, Selwyn Inlet dyke swarm (52.9°N, 131.8°W)							
Normal directions	12	0.3	76.5	62	5.6	—	—
Reversed directions	11	257.6	−75.9	29	8.6	—	—
All directions	23	37.6	79.2	29	5.8	—	—
VGP's	23	—	—	10	10.0	67.1	259.4
TS, Tasu Sound dyke swarm (52.8°N, 132.0°W)							
Normal directions	7	319.3	83.0	73	7.1	—	—
Reversed directions	9	166.9	−74.5	77	5.9	—	—
All directions	16	339.8	78.5	61	4.7	—	—
VGP's	16	—	—	20	8.5	71.7	204.7
LI, Lyell Island (including Bigsby Inlet) dyke swarm (52.7°N, 131.7°W)							
Normal directions	8	353.9	81.1	44	8.5	—	—
Reversed directions	10	173.2	−80.2	48	7.1	—	—
All directions	18	353.5	80.6	49	5.0	—	—
VGP's	18	—	—	15	9.2	70.3	222.0
BI, Burnaby Island dyke swarm (52.4°N, 131.3°W)							
Normal directions	2	17.1	69.3	91	26.5	—	—
Reversed directions	15	150.9	−81.0	24	8.0	—	—
All directions	17	341.4	80.2	24	7.4	—	—
VGP's	17	—	—	9	12.3	68.0	215.1
CB, Collison Bay dyke swarm (52.3°N, 131.2°W)							
Normal directions	10	337.3	77.9	58	6.4	—	—
Reversed directions	8	177.0	−77.1	114	5.2	—	—
All directions	18	346.4	77.7	74	4.0	—	—
VGP's	18	—	—	25	7.1	73.6	210.9
RH, Rose Harbour dyke swarm (52.2°N, 131.1°W)							
Normal directions	5	257.1	81.8	28	14.8	—	—
Reversed directions	11	159.8	−77.4	42	7.1	—	—
All directions	16	324.0	80.6	31	6.7	—	—
VGP's	16	—	—	10	12.0	64.0	203.7
MI, Masset Island Volcanics (53.6°N, 132.5°W)							
Normal directions	4	54.2	80.2	96	9.4	—	—
Reversed directions	4	205.3	−81.8	99	9.3	—	—
All directions	8	41.1	81.3	102	5.5	—	—
All VGP's	8	—	—	29	10.4	64.3	252.9
LP, Louise Island Pluton (53.0°N, 131.9°W)							
Normal directions	2	342.9	79.1	624	10.1	—	—
Reversed directions	3	153.3	−80.5	232	8.1	—	—
All directions	5	337.7	80.1	377	3.9	—	—
All VGP's	5	—	—	108	7.4	69.6	207.3

NOTES: B, number of dykes or sampling sites in volcanics and the pluton; λN and φE, latitude and longitude of paleopoles calculated as an average of virtual geomagnetic poles (VGP's) for each dyke; other headings as in Table 3. Directions given with respect to present horizontal. Mean directions are given for normal and reversed and for all directions irrespective of sign. For reason given in the notes to Table 3 the summary values for MI and LP differ slightly from those given in Hicken and Irving (1977).

spread extension in different directions at different times during the interval Late Eocene to Mid-Miocene. Paleomagnetically determined tilts carry with them implications regarding Tertiary stress fields, and the intriguing possibility arises that they may be used to complement the record of strain determined from the morphology of dyke swarms.

In Fig. 18 the simplest model compatible with our data is given. It is proposed that there was east−west extension during emplacement of the older north-trending Eocene dykes (Fig. 18*a*) followed by north−south extension during emplacement of the Miocene east-trending Selwyn Inlet dyke swarm (Fig. 18*b*). We correlate the latter phase of extension with the north−south extension recorded paleomagnetically by the northerly apparent tilts in older dykes (Fig. 17). Finally, it is suggested that the paleomagnetism of the Selwyn Inlet swarm and the lavas of Masset Inlet, both of Miocene age, record a third episode of tilting, to the northwest (Fig. 18*c*), implying extension in a north-northeast to south-southwest direction.

TABLE 5. Paleopoles from Middle Eocene through Miocene rock units in Canada and the U.S.A.

Entry	Rock unit	Age (Ma)	B	λN (°)	φE (°)	A_{95} (°)
1	Chilcotin Group basalts	12	48	84.0	219.7	5.1
2	Mojave–Sonora volcanics	10–20	65[a]	85.5	108.9	4.4
3	Columbia River basalts	15	59	88.7	171.6	4.0
4	Great Basin Ash flows	28	14	80.5	120.5	12.1
5	San Juan volcanics	29	36	85.0	114.0	7.5
6	Mogollan–Datil volcanics	31	61	81.5	147.3	2.4
7	Mistastin impact structure	39	10	85.5	117.7	3.0
8	Bearpaw Mountains volcanics	48	30	82.0	188.9	6.2
9	Absaroka volcanics	48	19	83.5	177.4	10.1
10	Highwood Mountains volcanics	50	29	82.1	163.0	7.0
11	Average of entries 1–5 (K = 271)	10–30	5	86.1	136.8	4.7
12	Average of entries 4–7 (K = 676)	20–40	4	83.3	127.0	3.5
13	Average of entries 6–10 (K = 564)	30–50	5	83.4	162.3	3.2
14	Average of entries 1–10 (K = 341)	10–50	10	84.9	152.8	2.6

NOTES: Age is that quoted in the original and has been determined generally from K–Ar studies. B, number of sampling sites; λN and φE, latitude and longitude of the paleopole; A_{95}, error at P = 0.05; K, fisher precision of paleopoles. The references are as follows: 1, Symons (1969); 2, Calderone *et al.* (1990); 3, Mankinen *et al.* (1987); 4, Grommé *et al.* (1972); 5, Beck *et al.* (1977); 6, Diehl *et al.* (1988); 7, Currie and Larochelle (1969); 8–10, Diehl *et al.* (1983). The above list gives the presently available paleopoles that are based on 10 or more collecting sites and have errors less than 15°.

[a]Number of cooling units.

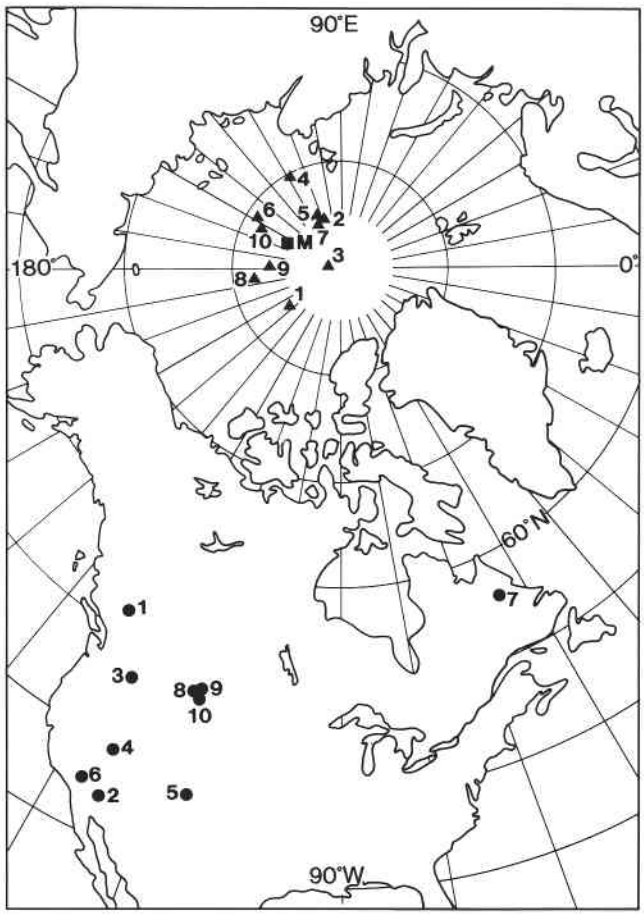


FIG. 15. Middle Eocene through Middle Miocene localities and paleopoles from tectonically undisturbed regions of Canada and the U.S.A. Numbering as in Table 5.

Phases *a* and *b* of Fig. 18 correspond to an interval of time during which plate reconstructions indicate that motion between the Pacific and North American plates ranged from oblique divergence to oblique convergence (Stock and Molnar 1988). The third phase of extension corresponds to a time when plate motions were strike-slip. It is emphasized that Fig. 18 is speculative, but that extension in varying directions occurred seems well established.

This interpretation is broadly consistent with the structural analysis of Lewis (1991), who recognized that the eastern side of the Louscoone Inlet fault (Fig. 1) has undergone extension and fragmentation into northerly tilted normal fault blocks. According to him the extension and normal faulting are linked to post-Early Oligocene dextral strike-slip motion, which may be related to phase *b* of Fig. 18. However, our data indicate that tilting is much more general than that invoked by Lewis, being present across the width of Moresby Island and extending north into Graham Island (Fig. 17). Also, the tilts recorded by him are less uniform and commonly greater than those we calculate, ranging up to 40°. These differences could reflect pre-Eocene tilts.

Except for the work of Lewis (1991), carried out over a limited area, the general northerly apparent tilts recorded paleomagnetically are not evident from geological mapping. Tilts of 10–15° are always difficult to estimate in regions of such sparse outcrop. Moreover, as already noted, there are difficulties in estimating tilt in lavas, and most dykes are orientated approximately north–south, which means that northerly tilt would not greatly change their attitude. In this situation it seems that paleomagnetism is the best general measure of original attitude.

It is possible that the two latest phases of extension (Fig. 18*b*, Fig. 18*c*) may have been related dynamically to the Queen Charlotte Island transform fault in a manner similar to that suggested by Diblee (1977) to explain tilting and extension

TABLE 6. Apparent tilts

Rock unit	D_m (°)	I_m (°)	t (Ma)	Preferred age range D_e (°)	Preferred age range I_c (°)	Preferred age range Tilt (°)	Full age range (10–50 Ma) D_e (°)	Full age range (10–50 Ma) I_e (°)	Full age range (10–50 Ma) Tilt (°)
SI, Selwyn Inlet swarm	37	79	10–30	354	69	15 at 323(11)	352	70	15 at 319
TS, Tasu Sound swarm	340	79	20–40	350	68	16 at 001(12)	352	70	10 at 006
LI, Lyell Island swarm	354	81	20–40	350	68	13 at 347(12)	352	70	11 at 352
BI, Burnaby Island swarm	341	80	30–50	350	70	13 at 356(13)	352	70	11 at 001
CB, Collison Bay swarm	346	78	30–50	350	70	10 at 352(13)	352	70	8 at 357
RH, Rose Harbour swarm	324	81	30–50	350	70	12 at 010(13)	352	70	13 at 011
MI, Masset Inlet volcanics	41	81	10–30	354	69	16 at 328(11)	352	71	16 at 326
LP, Louise Island pluton	338	81	30–50	349	71	09 at 004(13)	352	71	11 at 006

NOTES: D_m, I_m, observed mean direction are from Table 5; D_e, I_e, expected mean direction for stated time interval t calculated from mean paleopole for mainland North America (Table 5). The standard error in D_e and I_e for the age range 30–10 Ma is 3°, and 2° for all other ranges, including the full range. Calculations for the full age range (50–10 Ma) have been made using entry 14 of Table 5. Preferred age ranges are keyed to entries 11–13 of Table 5, as signified by number in parentheses.

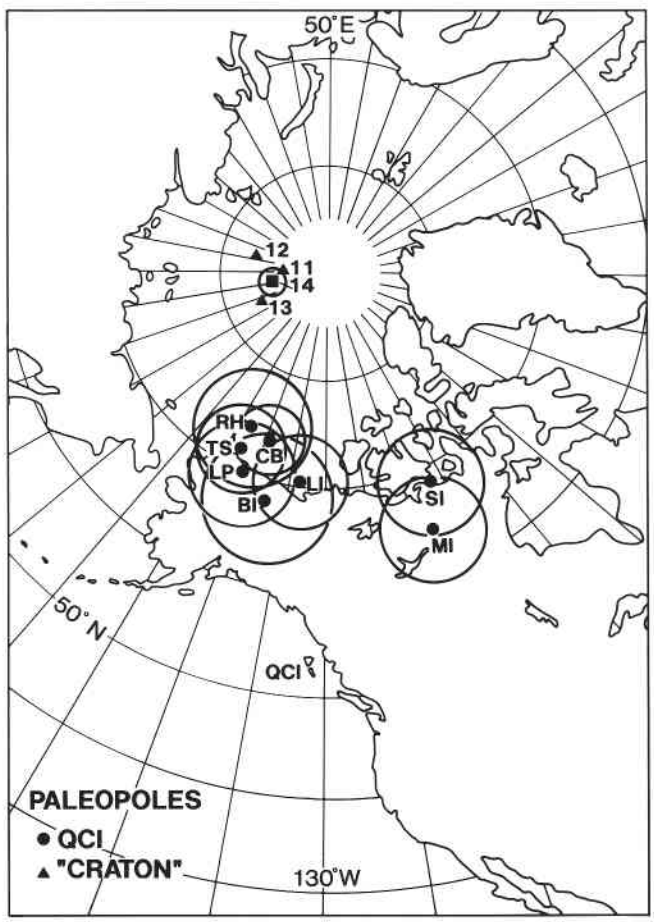


FIG. 16. Paleopoles from Queen Charlotte Islands compared with mean paleopoles from tectonically undisturbed regions of Canada and U.S.A. The latter numbered as in Table 5, the former lettered as in Table 6. Standard errors (P = 0.37) are drawn.

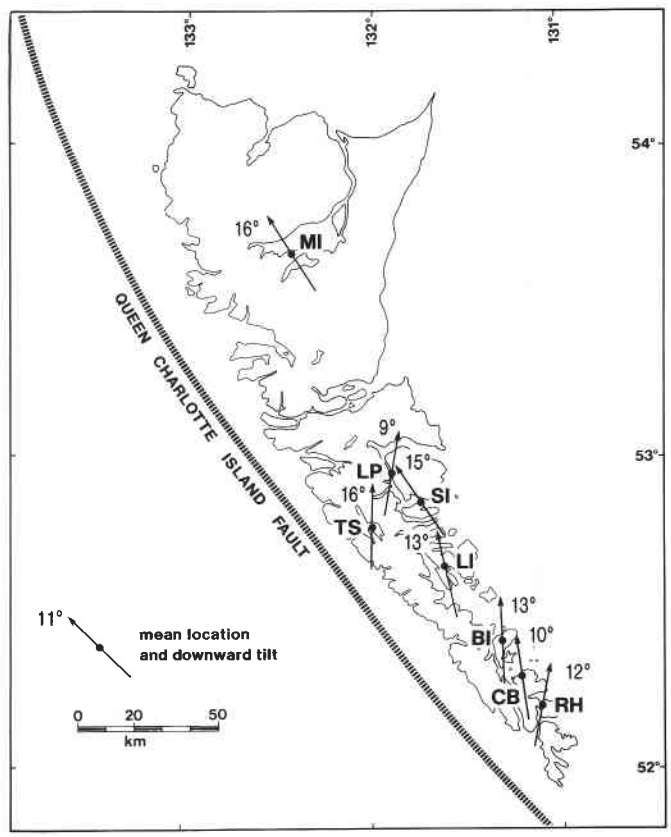


FIG. 17. Apparent tilts inferred from magnetization of rock-units lettered as in Table 6. Arrows denote the direction of downward tilt, and the number at each arrow head is the magnitude of tilt.

adjacent to the San Andreas Fault. The extension recorded by dyke orientations and by paleomagnetic data could be the result of block faulting and pull-apart structures in a complex transfer zone between the Queen Charlotte transform fault and the stable continental margin, as shown schematically in Fig. 19.

This is reminiscent of some interpretations of the evolution of the Gulf of California, for which two stages of extension have been invoked (Luhr *et al.* 1985; Allan *et al.* 1989). If this is correct, then the north–south extension of the second phase (Fig. 18*b*) presumably relates to a time when the Queen Charlotte transform had a more northerly trend, whereas phase *a* relates to extension that occurred when the transform had approximately its present orientation. Extending blocks probably were cut by longitudinal faults along which a small

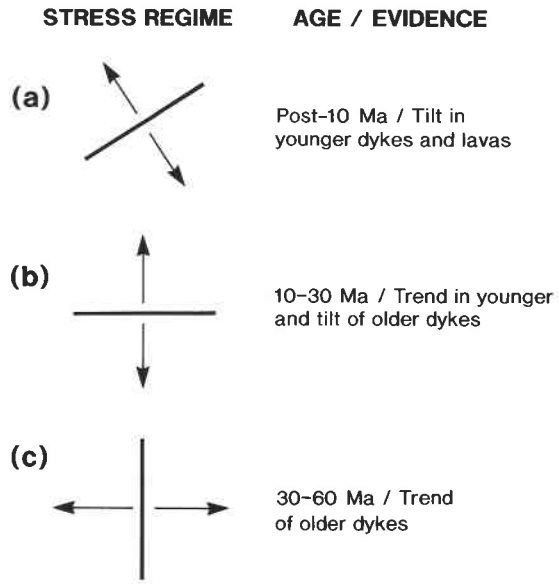


FIG. 18. Proposed variation of stress field with time. North is upward.

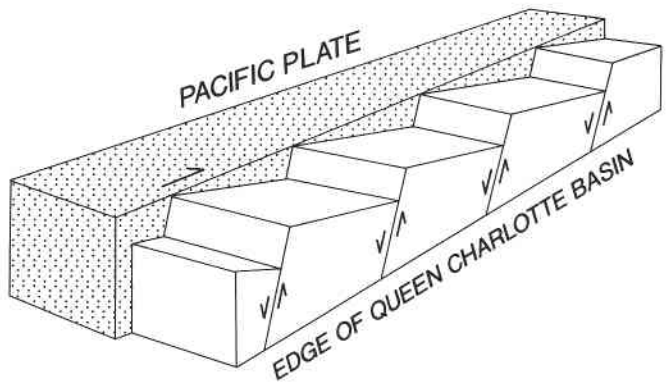


FIG. 19. Cartoon illustrating possible relationship of Queen Charlotte Island transform to extension of the western margin of Queen Charlotte Basin.

proportion of the displacement along the transform itself would be taken up. The Looscoone Inlet Fault may be such a fault related to phase *b* of Fig. 18.

As noted in the introduction, an extensional origin for the Queen Charlotte Basin is supported by many lines of evidence. The data of Figs. 14–17 and interpretation of Figs. 18 and 19 further support this hypothesis and provide information about the evolution of the stress field with time along the basin's western margin.

Systematic crustal tilts more widespread than previously supposed?

Paleomagnetic data from bedded Eocene rocks that are concordant with data from cratonic North America have now been obtained from across the Cordillera in Canada and adjacent U.S.A., implying that the major elements of the Canadian Cordillera were all in their present relative latitudinal position by Mid-Eocene time (Fox and Beck 1985; Bardoux and Irving 1989; Symons and Welling 1989; Bardoux and Irving 1989; Irving and Brandon 1990; Vandall and Palmer 1990). Because of this, aberrant magnetizations in Eocene and younger rocks can be interpreted as being caused by tilting or by rotations. In this way, systematic westerly tilts of 28–40° have been determined in the Eocene extension zone of southeastern British Columbia over a distance of 30 km (Irving and Archibald 1990; Marquis and Irving 1990). These tilts are related to regional basin and range style extension in a west to east direction (Parrish *et al.* 1988). Their systematic nature is presumably due to tilting in the hanging wall of very large north-trending extension faults. The present work implies systematic tilting of lesser magnitude over distances of about 200 km, activated, we suggest, by long continued motion along a nearby transform plate boundary.

These results are relevant to the discussion of the aberrant magnetization that have been observed in intrusions of mid-Cretaceous age in the Coast and Intermontane belts of the Cordillera over a region roughly 400 by 200 km (Symons 1974; Monger and Irving 1980; Beck *et al.* 1981; Irving *et al.* 1985). Such aberrations can be explained by assuming systematic tilting down to the southwest by about 30°, by translation of 2000 km from the south, or by some combination of the two. Initially there was some reluctance to accept the first because it was considered unlikely that tilt-producing processes could operate in such a uniform manner over such large distances (Beck 1980; Irving *et al.* 1985). The evidence cited above and described herein diminishes this argument because it now seems probable that, whatever their cause, systematic crustal tilts over distances of the order of tens and hundreds of kilometres have in fact occurred.

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